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GRADE 12 PHYSICS — REVISION BOOK 1

Peak Achievers Online Academy

Prepared by Hezekiah Omweri Nyatuka

Bachelor of Education (Science) — Chemistry & Physics

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Dedication

This book is dedicated to the loving memory of my late father, Simeon Nyatuka, whose guidance, sacrifices and belief in education continue to inspire my journey and the work I do as an educator.

To my mother, Hyrine Kerubo, for her love, strength and unwavering support throughout my life.

To my guardians, Peter Munde Mong'are and Teresa Kwamboka Sawe, for their guidance, encouragement and support.

To my friends, Mnaree, Richie, Duke, Caleb, Cuba, Philo, Rob, Tim, Mr Peter Kimani, and many others, whose friendship and encouragement have been part of my journey.

To Her, my special one, for being a special part of my journey and for the place you hold in my heart. ❤️

To my fellow teachers, whose commitment to education continues to inspire the work of helping learners succeed.

And to every learner who opens this book: may your curiosity grow, your confidence strengthen and your effort lead you towards excellence.

To each of you, this book carries a part of your contribution to my journey.

— Hezekiah Omweri Nyatuka

Preface

This book is designed for focused Grade 12 Physics revision. Each sub-strand moves from a concise explanation of the core idea to equations, worked examples, applications, investigations and

Contents

STRAND 1 — Mechanics and Thermal Physics

1.1 Dynamics

1.2 Uniform Circular Motion

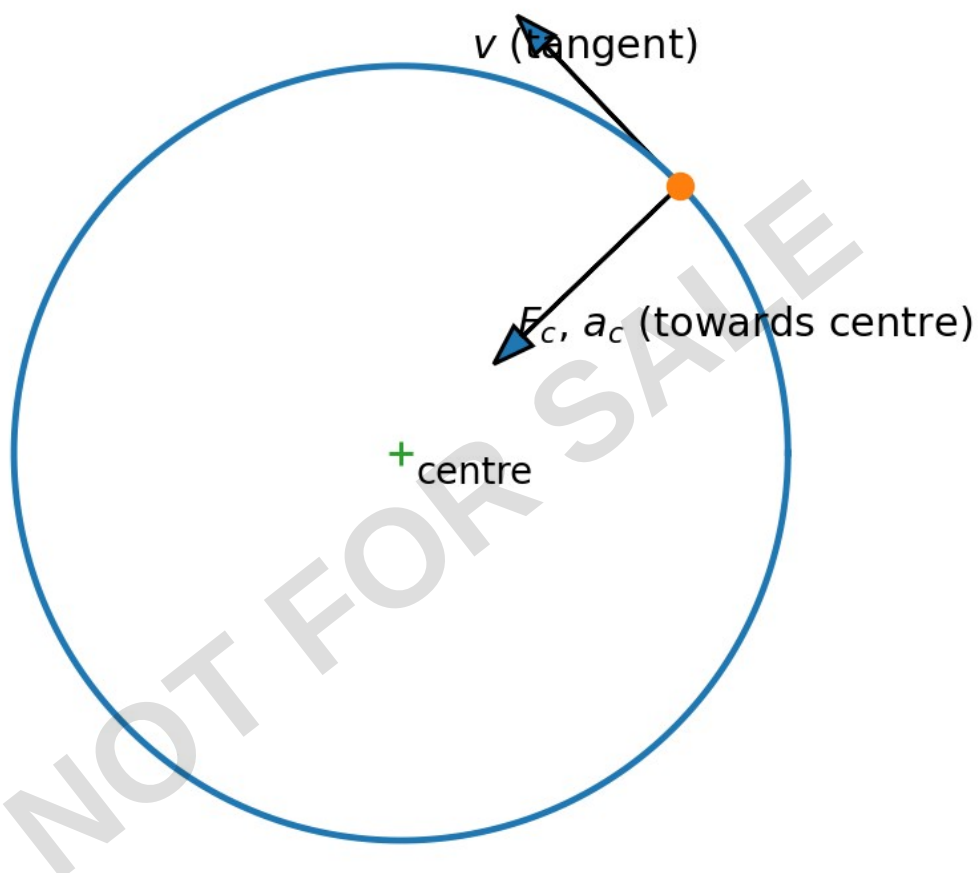


Figure: Velocity is tangential while centripetal force and acceleration act towards the centre.

1.3 Floating and Sinking

1.1 Dynamics

Dynamics deals with the relationship between force, motion and momentum. Momentum is a vector quantity, so direction must be considered.

Core Ideas

- Momentum is the product of mass and velocity.
- Impulse equals the change in momentum.
- For an isolated system, total linear momentum is conserved.
- Increasing stopping time reduces average force for the same change in momentum.

Key Equations

$$\mathbf{p} = m\mathbf{v}$$

$$\Delta\mathbf{p} = m(\mathbf{v} - \mathbf{u})$$

$$\mathbf{F} = \Delta\mathbf{p}/\Delta t = m\mathbf{a}$$

$$\mathbf{J} = \mathbf{F}\Delta t = \Delta\mathbf{p}$$

$$\Sigma\mathbf{p} \text{ before} = \Sigma\mathbf{p} \text{ after}$$

Worked Examples

Example 1

A 1200 kg car moves at 15 m/s. Find its momentum.

Solution: $\mathbf{p} = m\mathbf{v} = 1200 \times 15 = 1.80 \times 10^4 \text{ kg m/s.}$

Example 2

A 0.20 kg ball changes speed from 5 m/s to 25 m/s in 0.040 s. Find the average force.

Solution: $\Delta\mathbf{p} = 0.20(25-5)=4.0 \text{ kg m/s; } \mathbf{F}=4.0/0.040=100 \text{ N.}$

Example 3

A 4 kg trolley moving at 6 m/s hits a stationary 2 kg trolley and they stick.

Solution: $\mathbf{v}=(4\times6)/(4+2)=4.0 \text{ m/s.}$

Applications

- Seat belts, airbags and crumple zones increase stopping time and reduce force.
- Recoil and propulsion illustrate conservation of momentum.
- Sports follow-through changes impulse and force-time behaviour.

Investigation

Aim: To investigate conservation of momentum using trolleys.

Apparatus: Two trolleys, track, masses and a suitable timing/motion system.

Procedure: Measure masses and velocities before and after a collision. Compare total momentum.

Conclusion: Within experimental uncertainty, total momentum is approximately conserved when external forces are negligible.

Precaution: Keep the track level and minimise friction.

Quick Practice

1. Define momentum.
2. A force of 80 N acts for 0.15 s. Find the impulse.
3. Explain why airbags reduce injury.
4. Two bodies stick after collision. State the equation used.
5. A 4 kg device ejects 0.20 kg at 30 m/s. Find recoil speed if initially at rest.

Topic Revision Test

6. Explain conservation of momentum in a collision.
7. Calculate the force on a 0.50 kg ball changing from 4 m/s to 16 m/s in 0.060 s.
8. Explain why a longer collision time reduces average force.

EXAM FOCUS: State the governing principle first, then apply it to the situation. In calculations, always include the correct SI unit.

1.2 Uniform Circular Motion

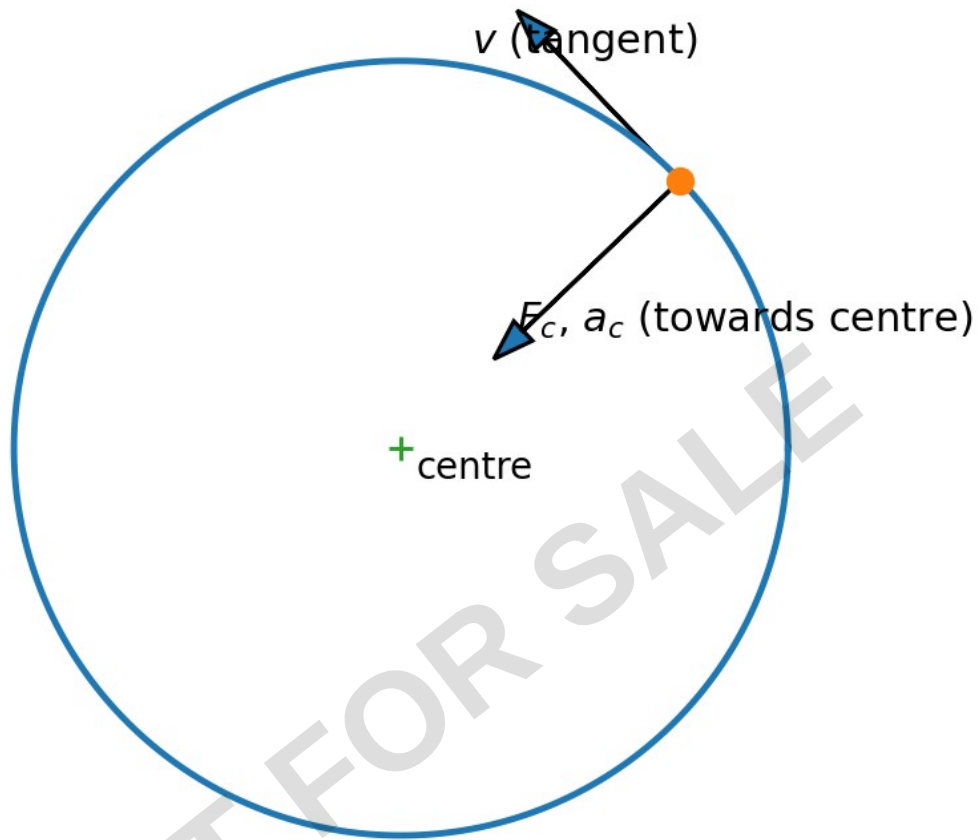


Figure: Velocity is tangential while centripetal force and acceleration act towards the centre.

Uniform circular motion is motion at constant speed along a circular path. The velocity changes continuously because its direction changes.

Core Ideas

- Velocity is tangential to the circular path.
- Centripetal acceleration and resultant force point toward the centre.
- Centripetal force is not a new force; it is the inward resultant provided by tension, friction, gravity or another interaction.
- If speed doubles at fixed r and m , centripetal force becomes four times as large.

Key Equations

$$\omega = \theta/t = 2\pi/T = 2\pi f$$

$$v = r\omega = 2\pi r/T$$

$$a_c = v^2/r = r\omega^2$$

$$F_c = mv^2/r = mr\omega^2$$

Worked Examples

Example 1

A body moves in a circle of radius 0.40 m with $\omega=5$ rad/s.

Solution: $v=r\omega=0.40\times5=2.0$ m/s.

Example 2

A car travels at 10 m/s around a bend of radius 50 m.

Solution: $a_c=v^2/r=100/50=2.0$ m/s².

Example 3

A 2 kg body moves at $\omega=3$ rad/s in a circle of radius 4 m.

Solution: $v=12$ m/s; $a_c=36$ m/s²; $F_c=72$ N.

Applications

- Road bends and vehicle safety.
- Centrifuges and rotating laboratory equipment.
- Satellite motion where gravity supplies the centripetal force.

Investigation

Aim: To relate circular speed, radius and inward force.

Apparatus: Rubber stopper or safe mass, string, tube, hanging masses and stopwatch.

Procedure: Rotate the mass steadily and compare changes when speed or radius is varied.

Observation: Greater speed requires a larger inward force for the same radius.

Precaution: Keep the rotating mass clear of people and objects.

Quick Practice

9. State the direction of centripetal acceleration.
10. Find v for $r=2$ m and $\omega=4$ rad/s.
11. Explain what happens to F_c if v doubles.
12. What path does a body follow if the string suddenly breaks?

Topic Revision Test

13. A 1.5 kg body moves at 6 m/s in a circle of radius 3 m. Find a_c and F_c .
14. Distinguish speed from velocity in circular motion.
15. Explain how gravity acts as centripetal force for a satellite.

EXAM FOCUS: State the governing principle first, then apply it to the situation. In calculations, always include the correct SI unit.